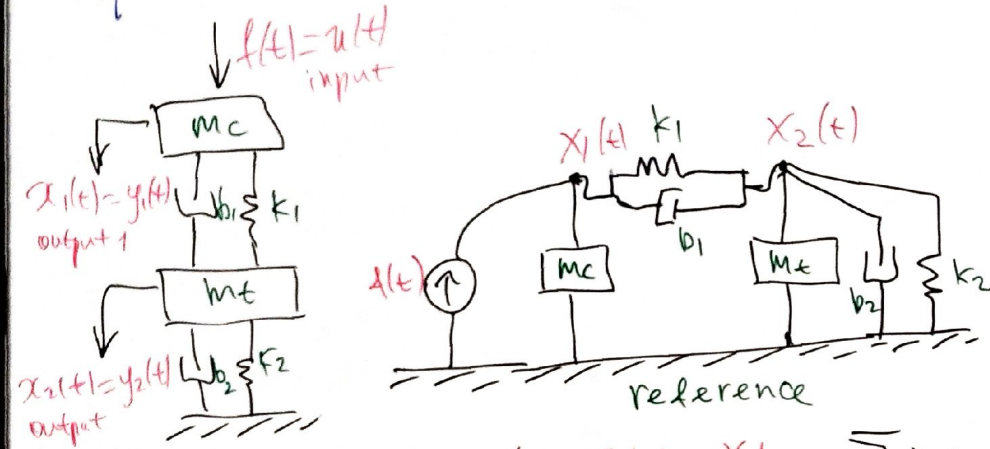


1) Translational system:



m_c - car mass
 m_t - tyre mass

$$G_1(s) = \frac{X_1(s)}{F(s)} - ?$$

$$G_2(s) = \frac{X_2(s)}{F(s)} - ?$$

At Node X_1 : $\sum \text{incoming forces} = \sum \text{outgoing forces}$

$$m_c \ddot{x}_1 + b_1(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_2) = f(t) \quad (*)$$

At Node X_2 :

$$m_t \ddot{x}_2 + b_2 \dot{x}_2 + k_2 x_2 + k_1(x_2 - x_1) + b_1(\dot{x}_2 - \dot{x}_1) = 0$$

Laplace Transform:

$$\begin{cases} (m_c s^2 + b_1 s + k_1) X_1(s) + (-b_1 s - k_1) X_2(s) = F(s) \\ (-b_1 s - k_1) X_1(s) + (m_t s^2 + (b_1 + b_2)s + k_1 + k_2) X_2(s) = 0 \end{cases}$$

метод Крамера:

$$X_1(s) = \frac{\Delta_1}{\Delta}$$

$$X_1(s) = \frac{\begin{vmatrix} b_1 & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}; \quad X_2(s) = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

$$X_1(s) = \frac{b_1 a_{22} - a_{12} a_{21}}{a_{11} a_{22} - a_{12} a_{21}}; \quad X_2(s) = \frac{a_{11} b_2 - b_1 a_{21}}{a_{11} a_{22} - a_{12} a_{21}}$$

$$G_1(s) = \frac{X_1(s)}{F(s)}; \quad X_1(s) = G_1(s) F(s); \quad X_2(s) = G_2(s) F(s)$$

$$\begin{bmatrix} m_c s^2 + b_1 s + k_1 & -b_1 s - k_1 \\ -b_1 s - k_1 & m_t s^2 + (b_1 + b_2)s + k_1 + k_2 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$A \cdot X = B$

$$A^{-1} \cdot A \cdot X = A^{-1} \cdot B$$

$$E \cdot X = A^{-1} \cdot B$$

$$X = A^{-1} \cdot B$$

translational: Force-Voltage:

$$\left. \begin{array}{l} m_c = 400 \\ m_t = 10 \\ b_1 = 1000 \\ k_1 = 2000 \\ b_2 = 2500 \\ k_2 = 8000 \end{array} \right\} \left. \begin{array}{l} L_c = 400 \\ L_t = 10 \\ R_1 = 1000 \\ C_1 = 1/k_1 = 1/2000 \\ R_2 = 2500 \\ C_2 = 1/k_2 = 1/8000 \end{array} \right\}$$

$$f(t) = 800(N) \rightarrow V(t) = 800(V)$$

$$I(t) = 800(A)$$